SOFTWARE BLUEPRINT · PHASE 2

Software Requirements

Numbered, testable requirements, use cases and state models for the proposed Usiri Marketplace

Working name only — no Usiri affiliation is implied.

Prepared for: founders, technical lead, design, security and test reviewers
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Date: 30 October 2026
Baseline: UM-BRD-001 v1.0

DRAFT FOR REVIEW. No implementation is authorized. Requirements marked Must form the launch scope. Values marked TBC come from configuration, not code.

CONFIDENTIAL — FOR INTERNAL DISCUSSION

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0. Document control

Field	Value
Document	Software Requirements Specification
Project	Usiri Marketplace (working name)
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Classification	Confidential — internal discussion
Baseline	UM-BRD-001 v1.0; UM-DISC-001 v0.2; UM-AMD-001 v1.0; UM-DEC-001 v0.2
Next documents	System architecture, database design, API specification, security architecture, UX specification

> **Status notice.** Usiri is a potential logistics and brand partner only. No API, rate card, franchise term or licence is assumed. The platform is carrier-agnostic and PSP-agnostic. No code has been written.

0.1 Conventions

- **Shall** marks a mandatory requirement. **Should** marks a requirement that may be deferred with a written decision. **May** marks an optional capability.
- Every requirement has a stable ID of the form `REQ-<module>-<number>`. IDs are never reused or renumbered.
- **Priority** uses MoSCoW: Must, Should, Could, Won't. **Must** requirements form the launch scope.
- **Source** links to a business need (BN), business rule (BR), capability (CAP) or decision (D).
- **Verification** states how the requirement will be proven. It is not a test script; it names the test approach.
- **TBC** marks a value that cannot be fixed until an open decision is answered. A TBC value must come from configuration, not from code.

1. Introduction

1.1 Purpose

This specification defines the software behaviour required to operate the marketplace described in the BRD. It is the contract between business, design, engineering and testing. It is detailed enough to estimate, build and test, and it deliberately stops short of prescribing the internal design.

1.2 Scope

The system comprises a customer web experience, a supplier portal, an operations console, a logistics coordination module, a payment and settlement module, a support and returns workflow, and the platform services that connect them. It runs as a modular monolith with a relational database, background workers, object storage and external integrations for payments, messaging and delivery.

1.3 Definitions

Term	Meaning
Parent order	The customer’s single commercial order created by one checkout
Supplier order	The part of a parent order fulfilled by one supplier; the unit of commission and settlement
Shipment	A physical parcel with a carrier, waybill and delivery evidence
Variant	A purchasable option combination with its own SKU, price and stock
Reservation	A temporary hold on stock while payment is in progress
Captured	Funds successfully taken by the PSP
Settled	Supplier payable released for payout after delivery and the return window
POD	Proof of delivery, with recipient evidence
Mode A	Manual carrier coordination
Mode B	Carrier integration through an adapter
TBC	To be confirmed by an open decision

1.4 References

UM-DISC-001 Requirements Discovery Report v0.2; UM-BRD-001 v1.0; UM-AMD-001 v1.0; UM-DEC-001 v0.2; the concept document v1.0; the startup workbook.

2. Overall description

2.1 Product perspective

The system is a new, self-contained platform. It integrates with, but does not depend on, a specific payment provider or carrier. It replaces spreadsheets and messaging apps for the operating team; it does not replace the supplier’s own shop or stock system, which remains the source of stock truth through portal updates or file upload.

2.2 Product functions

1. Customer identity, consent and addresses.
2. Supplier onboarding, verification and contracts.
3. Catalogue, variants, attributes, media and moderation.
4. Inventory and reservations.

5. Search, browse and product discovery.
6. Multi-supplier cart and checkout.
7. Order orchestration and partial fulfilment.
8. Payment initiation, verification, refunds and reconciliation.
9. Commission, statements, payouts and holds.
10. Logistics coordination, tracking, proof of delivery and claims.
11. Category-sensitive returns and disputes.
12. Customer support with order context.
13. Transactional and consent-aware notifications.
14. Administration, access control, configuration and audit.
15. Reporting and supplier scorecards.
16. Verified reviews, controlled promotions and configurable brand rules.

2.3 User classes

Class	Frequency	Criticality	Notes
Guest	High	High	Must complete checkout without an account
Customer	High	High	Mobile, low bandwidth, Swahili-first
Supplier owner	Daily	High	May use a phone; needs simple tasks
Supplier staff	Daily	Medium	Packing and dispatch only
Support agent	Daily	High	Needs order context in one screen
Moderator	Daily	High	Queue-based work
Finance	Daily	High	Reconciliation and payouts
Operations lead	Daily	High	Exceptions and carrier coordination
Administrator	Weekly	High	Access and configuration
Carrier administrator	Daily	Medium	Assigned shipments only
Auditor or founder	Monthly	Medium	Read-only access to evidence

2.4 Operating environment

Mobile web first, responsive to desktop. Supported browsers: current Chrome, Safari, Firefox and Android WebView. The customer experience must work on a mid-range Android device on a 4G connection, and degrade gracefully on slower devices. The supplier portal must be usable on a phone. The operations console is desktop-first. No native application is in scope.

2.5 Design and implementation constraints

- Modular monolith; a single deployable application with clear module boundaries.
- PostgreSQL as the system of record; Redis for ephemeral state such as sessions, rate limits and reservations.
- Object storage for media and documents, with signed access.
- All money as integer TZS.
- Server-side calculation of all prices, fees, commissions and refunds.

- No card data stored or transmitted through platform infrastructure beyond the PSP’s hosted interface.
- All customer-facing strings externalised for Swahili and English.
- All brand, territory and franchise behaviour driven by configuration.
- Timezone Africa/Dar_es_Salaam for business dates; UTC for storage.

2.6 Assumptions and dependencies

The PSP provides signed callbacks, a status endpoint, refunds and a settlement or payout mechanism. A licensed carrier can carry the launch categories and provide a waybill and delivery evidence. SMS delivery is available to Tanzanian numbers. The team can obtain legal advice before launch. These are dependencies, not confirmed facts.

3. System context and actors

Actor	Interaction	Trust boundary
Guest or customer	Web application over HTTPS	Untrusted input; authenticated for account actions
Supplier owner or staff	Supplier portal	Authenticated; scoped to one supplier
Carrier administrator or agent	Carrier view or manual coordination	Authenticated; scoped to assigned shipments
Support, moderator, finance, operations	Operations console	Authenticated with MFA; role-scoped
PSP	Server-to-server API and signed callbacks	External; verified signatures and IP allow-list where available
Carrier	Adapter or manual entry	External; contract and evidence
Messaging provider	Server-to-server API	External; template and consent rules
Object storage	Private bucket with signed URLs	Internal; least privilege
Auditor	Read-only console and exports	Authenticated; no mutation

Data flows across these boundaries are described in Section 7. The most sensitive boundaries are the PSP callback, supplier access to customer delivery data, and staff access to KYC documents.

4. Functional requirements

4.1 Identity, accounts and consent (REQ-ACC)

ID	Requirement	Priority	Source	Verification
REQ-ACC-001	The system shall let a customer register and sign in with a phone number verified by a one-time code.	Must	CAP-01	OTP integration test for valid, expired and reused codes
REQ-ACC-002	The system shall allow guest checkout with only the contact and address data needed to fulfil the order.	Must	BN-01, STK-01	Guest order completes without an account
REQ-ACC-003	The system shall offer account creation after a guest order without requiring re-entry of the order data.	Should	CAP-01	Conversion test from guest order to account

ID	Requirement	Priority	Source	Verification
REQ-ACC-004	The system shall maintain a customer profile with name, phone, optional email and a list of addresses with a landmark field.	Must	CAP-01	Create, edit and select address in checkout
REQ-ACC-005	The system shall record consent per purpose against a versioned privacy notice, with timestamp and source.	Must	BR-024, NFR-PRIV-01	Consent record inspection for each signup path
REQ-ACC-006	The system shall expire idle sessions and invalidate sessions on sign-out and on credential change.	Must	NFR-SEC-01	Session tests
REQ-ACC-007	The system shall require multi-factor authentication for all staff and supplier owner accounts.	Must	NFR-SEC-03	Login test for each privileged role
REQ-ACC-008	The system shall store passwords, where used, with Argon2id and enforce a minimum length of 12 characters.	Must	NFR-SEC-03	Code review and authentication test
REQ-ACC-009	The system shall let a customer export their personal data in a machine-readable format.	Must	NFR-PRIV-03	Export request produces complete file
REQ-ACC-010	The system shall support account deletion or anonymisation while retaining records required for tax, dispute and audit.	Must	BR-024	Deletion test with financial retention check
REQ-ACC-011	The system shall keep marketing consent separate from transactional messaging and allow withdrawal at any time.	Must	BR-025	Withdrawal stops marketing only
REQ-ACC-012	The system shall rate-limit sign-in, OTP and password reset attempts and lock progressively.	Must	NFR-SEC-05	Abuse test against limits

4.2 Supplier onboarding and KYC (REQ-KYC)

ID	Requirement	Priority	Source	Verification
REQ-KYC-001	The system shall accept a supplier application with business name, trading name, category, location, contact and owner details.	Must	CAP-02	Application submitted and stored
REQ-KYC-002	The system shall accept document uploads for business registration, TIN, owner identity, shop evidence and payout account evidence.	Must	CAP-02, REG-07	Upload each document type with size and type checks
REQ-KYC-003	The system shall move an application through defined states: submitted, under review, information requested, approved, rejected, suspended.	Must	CAP-02	State transition test with reasons
REQ-KYC-004	The system shall detect possible duplicate suppliers by TIN, phone number and payout account.	Should	CAP-02	Duplicate application flagged
REQ-KYC-005	The system shall record a site visit with date, visitor, location evidence and outcome.	Must	CAP-02	Site visit recorded before approval
REQ-KYC-006	The system shall store the supplier contract version accepted, with date and signatory.	Must	REG-08	Contract record linked to supplier
REQ-KYC-007	The system shall require a reason and an actor for every approval, rejection and suspension decision.	Must	BR-023	Decision without reason is blocked
REQ-KYC-008	The system shall verify that the payout account name matches the supplier’s legal or trading name before approval.	Must	BR-029	Mismatch blocks approval pending review
REQ-KYC-009	The system shall schedule re-verification and flag overdue verifications, ownership changes and bank-account changes.	Must	CAP-02	Overdue supplier appears in compliance queue
REQ-KYC-010	The system shall let an authorised operator suspend a supplier, immediately hiding its listings and blocking new orders.	Must	BR-010	Suspension test shows listings hidden
REQ-KYC-011	The system shall let a supplier owner invite staff with limited permissions and revoke access immediately.	Must	ROL-04	Invitation and revocation test
REQ-KYC-012	The system shall notify the supplier of missing or expiring documents in the supplier’s chosen language.	Should	CAP-13	Notification triggered at threshold

ID	Requirement	Priority	Source	Verification
REQ-KYC-013	The system shall allow a rejected supplier to correct and resubmit without creating a duplicate record.	Should	CAP-02	Resubmission test
REQ-KYC-014	The system shall publish a supplier storefront with approved brand, description, policies and live listings only.	Must	BN-03	Storefront shows approved content only

4.3 Catalogue, variants and moderation (REQ-CAT)

ID	Requirement	Priority	Source	Verification
REQ-CAT-001	The system shall maintain a category tree of six departments with configurable subcategories.	Must	BN-03, D-13	Category administration test
REQ-CAT-002	The system shall support category attribute templates with mandatory and optional fields, data types and allowed values.	Must	BR-034	Listing cannot publish without mandatory attributes
REQ-CAT-003	The system shall support products with one or more variants defined by option types such as size, colour, scent, pack size and model.	Must	BR-033	Variant creation and purchase test
REQ-CAT-004	The system shall require a unique SKU per variant and reject duplicates within a supplier account.	Must	BR-033	Duplicate SKU rejected
REQ-CAT-005	The system shall store price and stock at variant level and display a price range when variants differ.	Must	BR-033	Price range and per-variant checkout
REQ-CAT-006	The system shall support multiple images per product and per variant, with ordering, alt text and category-specific guidance.	Must	BN-03	Media upload and ordering test
REQ-CAT-007	The system shall route new and edited listings to a moderation queue with the applicable category checklist.	Must	CAP-03	Listing enters correct queue
REQ-CAT-008	The system shall let a moderator approve, reject or request changes with a reason code and a note.	Must	BR-010	Decision recorded with reason
REQ-CAT-009	The system shall prevent publication of a listing that is unapproved, out of stock, prohibited or missing mandatory attributes.	Must	BR-010, BR-015	Publish attempt is blocked with explanation
REQ-CAT-010	The system shall support bulk listing upload by CSV with per-row validation and an error report.	Should	CAP-03	200-row upload with mixed errors
REQ-CAT-011	The system shall maintain a brand registry and flag listings that claim a brand without evidence where evidence is required.	Should	BR-036	Brand claim routed to evidence check
REQ-CAT-012	The system shall block prohibited items at listing time using a configurable keyword, category and brand list.	Must	BR-015	Prohibited term blocked
REQ-CAT-013	The system shall store category size guides and link them from fashion, baby and footwear listings.	Must	BN-03, D-18	Size guide visible on product page
REQ-CAT-014	The system shall support periodic audit of live listings by sampling or filter, with findings and corrective actions.	Should	CAP-03	Audit sample created and recorded
REQ-CAT-015	The system shall provide category-specific moderation checklists editable without deployment.	Must	BN-16	Checklist change reflected in queue
REQ-CAT-016	The system shall keep a version history of listing content, price and moderation decisions.	Must	BR-011	Prior version retrievable

4.4 Inventory and availability (REQ-INV)

ID	Requirement	Priority	Source	Verification
REQ-INV-001	The system shall record stock per variant per supplier location.	Must	CAP-04	Stock record created and read
REQ-INV-002	The system shall reserve stock atomically when a payment intent is created and release it if payment fails or the reservation expires.	Must	BR-002	Concurrent checkout test shows no oversell

ID	Requirement	Priority	Source	Verification
REQ-INV-003	The system shall apply a configurable reservation expiry, defaulting to 15 minutes.	Must	BR-002	Expired reservation releases stock
REQ-INV-004	The system shall prevent negative available stock through database-level constraints or equivalent guarantees.	Must	BN-05	Concurrent reservation test
REQ-INV-005	The system shall hide or mark unavailable a variant at zero available stock.	Must	BR-010	Zero-stock variant not purchasable
REQ-INV-006	The system shall alert a supplier when stock falls below a configurable threshold.	Should	CAP-04	Threshold alert sent
REQ-INV-007	The system shall let a supplier update stock by portal edit or file upload with an audit record.	Must	CAP-04	Update and audit entry
REQ-INV-008	The system shall record manual stock adjustments with actor, reason and before-and-after values.	Must	BR-023	Adjustment history visible
REQ-INV-009	The system shall not accept backorders at launch.	Must	D-02	Order attempt beyond stock is blocked

4.5 Search, browse and discovery (REQ-DIS)

ID	Requirement	Priority	Source	Verification
REQ-DIS-001	The system shall provide keyword search across product title, brand, category and attributes in Swahili and English.	Must	CAP-05	Search test in both languages
REQ-DIS-002	The system shall support filters by category, price, availability, brand and category-specific attributes.	Must	CAP-05	Filter combinations return correct sets
REQ-DIS-003	The system shall support sorting by relevance, price and recency.	Must	CAP-05	Sort order test
REQ-DIS-004	The system shall provide department and category landing pages with curated ordering.	Must	CAP-05	Landing page renders live products
REQ-DIS-005	The system shall provide a product page with variant selection, price, stock, delivery estimate, return terms and supplier identity.	Must	BN-06	Product page completeness check
REQ-DIS-006	The system shall handle zero results with spelling suggestions, related categories and popular products.	Should	CAP-05	Zero-result page shows alternatives
REQ-DIS-007	The system shall show related products from the same category without promoting unrelated categories.	Could	CAP-05	Related list relevance check
REQ-DIS-008	The system shall present all customer-facing discovery content in the selected language.	Must	NFR-I18N-01	Pseudo-localisation test
REQ-DIS-009	The system shall record search terms and zero-result queries for merchandising review.	Should	CAP-15	Search report available
REQ-DIS-010	The system shall indicate when a product is unavailable rather than hiding it abruptly from a shared link.	Should	BN-05	Shared link shows unavailable state

4.6 Cart and checkout (REQ-CHK)

ID	Requirement	Priority	Source	Verification
REQ-CHK-001	The system shall maintain one cart that may contain items from multiple suppliers.	Must	BN-01	Three-supplier cart
REQ-CHK-002	The system shall validate price and stock server-side at cart view, checkout start and order submission.	Must	BR-003	Tampering and stale-price tests
REQ-CHK-003	The system shall collect a delivery address with recipient name, phone, street or area, landmark and optional instructions.	Must	CAP-06	Address validation test
REQ-CHK-004	The system shall quote delivery per supplier shipment and show the number of parcels before payment.	Must	BN-06	Multi-supplier quote displayed

ID	Requirement	Priority	Source	Verification
REQ-CHK-005	The system shall display item value, delivery, discounts, any small-order fee and total in TZS before payment.	Must	BN-06, BR-038	Total breakdown visible and correct
REQ-CHK-006	The system shall enforce a configurable minimum delivered order value or small-order fee, disclosed before payment.	Must	BR-038, D-14	Below-threshold cart shows fee or block
REQ-CHK-007	The system shall require acceptance of terms, returns policy and privacy notice before order submission.	Must	REG-02, BR-024	Consent recorded with version
REQ-CHK-008	The system shall support voucher or promotion codes where a valid campaign exists.	Should	CAP-17	Valid and invalid code tests
REQ-CHK-009	The system shall submit an order idempotently so that retries and double taps do not create duplicate orders.	Must	NFR-R-01	Double-submit test creates one order
REQ-CHK-010	The system shall create stock reservations before payment initiation and release them on failure or expiry.	Must	BR-002	Reservation lifecycle test
REQ-CHK-011	The system shall allow a guest to provide only the contact details required for fulfilment and support.	Must	STK-01	Guest checkout field audit
REQ-CHK-012	The system shall show an estimated delivery window per supplier based on configured handling and transit times.	Should	BN-06	Estimate shown and labelled as estimate
REQ-CHK-013	The system shall preserve cart contents across sessions for a signed-in customer for a configurable period.	Could	CAP-06	Cart restored after sign-in
REQ-CHK-014	The system shall recover gracefully from payment failure, preserving the cart and offering retry.	Must	BN-07	Failed payment returns to cart with items
REQ-CHK-015	The system shall show the supplier name and fulfilment responsibility on each cart line.	Must	BN-01	Supplier shown per line

4.7 Orders and fulfilment orchestration (REQ-ORD)

ID	Requirement	Priority	Source	Verification
REQ-ORD-001	The system shall create one parent order and one supplier order per supplier at checkout.	Must	BR-001	Three-supplier order structure
REQ-ORD-002	The system shall give the supplier a configurable acceptance window, defaulting to 12 working hours, after which the supplier order may be auto-cancelled.	Must	CAP-07	Timeout test
REQ-ORD-003	The system shall allow a supplier to accept, reject or partially accept an order with a reason.	Must	BN-09	Partial acceptance test
REQ-ORD-004	The system shall allow cancellation of a supplier order before acceptance.	Must	BR-012	Cancellation before acceptance
REQ-ORD-005	The system shall allow cancellation of individual lines after acceptance only through the return or exception process.	Must	BR-012	Post-acceptance cancel blocked
REQ-ORD-006	The system shall maintain separate payment, supplier order, shipment, return and dispute states and derive a customer-facing summary.	Must	CAP-07	State combination tests
REQ-ORD-007	The system shall record every state change with actor, timestamp, reason and previous state.	Must	BR-023	Event history inspection
REQ-ORD-008	The system shall show the customer a per-supplier progress view within the parent order.	Must	BN-01	Customer view test
REQ-ORD-009	The system shall show the supplier only its own order lines and the customer data needed to fulfil.	Must	BR-013	Supplier data visibility test
REQ-ORD-010	The system shall issue a receipt or invoice per parent order and a supplier statement per settlement period.	Must	REG-06	Document generation test
REQ-ORD-011	The system shall allow a customer to reorder previous items subject to current price and stock.	Could	CAP-07	Reorder test with changed price
REQ-ORD-012	The system shall support order search by order number, phone, supplier and tracking reference.	Must	NFR-SUP-01	Lookup under 30 seconds

ID	Requirement	Priority	Source	Verification
REQ-ORD-013	The system shall notify the customer of acceptance, dispatch, delivery and cancellation events.	Must	CAP-13	Notification trigger test
REQ-ORD-014	The system shall calculate and display the expected supplier payout date on the supplier order.	Should	BR-007	Payout date matches policy

4.8 Payments, refunds and reconciliation (REQ-PAY)

ID	Requirement	Priority	Source	Verification
REQ-PAY-001	The system shall initiate payment through the PSP for the parent order total and store the PSP reference.	Must	CAP-08	Sandbox initiation
REQ-PAY-002	The system shall verify every PSP callback signature and reject unsigned or invalid callbacks.	Must	NFR-SEC-01	Forged callback rejected
REQ-PAY-003	The system shall confirm final payment status through a server-to-server PSP query before marking an order paid.	Must	BN-07	Callback followed by status query
REQ-PAY-004	The system shall process duplicate callbacks idempotently.	Must	NFR-R-01	Repeated callback leaves one payment
REQ-PAY-005	The system shall record failed, cancelled, expired and reversed payments with a reason.	Must	BN-07	Failure state test
REQ-PAY-006	The system shall support full and partial refunds against the original payment with a reason and an approver.	Must	BR-027	Partial refund test
REQ-PAY-007	The system shall prevent refunds exceeding the captured amount less prior refunds and reversals.	Must	BR-027	Over-refund blocked
REQ-PAY-008	The system shall record chargebacks and reversals with evidence and affected order lines.	Must	RSK-13	Chargeback record created
REQ-PAY-009	The system shall post every payment, refund and fee to a double-entry ledger with unique references.	Must	BN-23	Ledger balances to zero per transaction
REQ-PAY-010	The system shall reconcile PSP settlement files to the ledger daily and queue unmatched items.	Must	BN-23	Reconciliation run with seeded mismatch
REQ-PAY-011	The system shall provide an exception queue for unmatched, duplicated or amount-mismatched transactions.	Must	BN-23	Exception appears and can be resolved
REQ-PAY-012	The system shall show the customer the payment status of each order without exposing PSP credentials or card data.	Must	BN-07	Status view test
REQ-PAY-013	The system shall not store card numbers, CVV or full track data in any platform system.	Must	NFR-PRIV-02	Data-inventory review
REQ-PAY-014	The system shall allow a customer to retry a failed payment while the reservation is valid.	Must	BN-07	Retry within window

4.9 Commission, settlement and payouts (REQ-SET)

ID	Requirement	Priority	Source	Verification
REQ-SET-001	The system shall calculate commission per order line on net completed merchandise using the effective rule at order time.	Must	BR-004	Calculation test with returns and discounts
REQ-SET-002	The system shall record the commission rule version on every order line.	Must	BR-011	Rule version visible on statement
REQ-SET-003	The system shall generate a supplier statement per period with orders, commission, refunds, chargebacks, adjustments and net payable.	Must	CAP-09	Statement reconciles to ledger
REQ-SET-004	The system shall make a payable eligible only after delivery plus the return window, unless a hold applies.	Must	BR-007	Eligibility date test

ID	Requirement	Priority	Source	Verification
REQ-SET-005	The system shall apply holds for open disputes, negative balances, KYC issues and risk flags with a written reason.	Must	BR-029	Hold blocks payout and is visible
REQ-SET-006	The system shall build payout batches and require a different approver before release.	Must	BR-028	Same-user approval blocked
REQ-SET-007	The system shall export a payout file in the format required by the payment channel.	Must	CAP-09	Export accepted by channel in test
REQ-SET-008	The system shall record manual adjustments with reason, actor and approver, and reflect them in the next statement.	Must	BR-009	Adjustment appears in statement
REQ-SET-009	The system shall carry a negative supplier balance forward and offset it against future payables.	Should	BR-029	Negative balance offset test
REQ-SET-010	The system shall apply configurable tax withholding where counsel confirms a requirement.	Should	REG-06	Withholding applied when enabled
REQ-SET-011	The system shall show the supplier the payout status and reference for each paid batch.	Must	BN-08	Supplier view test
REQ-SET-012	The system shall reconcile payout channel confirmations to the payout batch and flag failures.	Must	BN-23	Failed payout flagged

4.10 Logistics and tracking (REQ-LOG)

ID	Requirement	Priority	Source	Verification
REQ-LOG-001	The system shall maintain a register of carriers with services, coverage, contact details and contract status.	Must	CAP-10	Carrier administration test
REQ-LOG-002	The system shall maintain delivery zones and rate rules by carrier, weight, dimensions and category.	Must	BN-06	Quote test for each zone and category
REQ-LOG-003	The system shall create a shipment per supplier order or parcel with items, weight, dimensions and declared value.	Must	BR-016	Shipment record test
REQ-LOG-004	The system shall record carrier, service, waybill, cost and pickup date for every shipment.	Must	BR-016	Required fields enforced
REQ-LOG-005	The system shall support a manual booking mode in which an operator records the booking and status events.	Must	BR-031	Mode A end-to-end test
REQ-LOG-006	The system shall define a carrier adapter interface so an integrated carrier can replace manual entry without changing order logic.	Must	BR-031, D-09	Adapter contract review and stub test
REQ-LOG-007	The system shall record shipment status events with source, timestamp and raw reference.	Must	BN-11	Event timeline test
REQ-LOG-008	The system shall record proof of delivery with recipient name, timestamp, method and evidence where available.	Must	BR-016	POD required before delivery completion
REQ-LOG-009	The system shall record failed delivery attempts with reason and schedule a retry per policy.	Must	BR-017	Failed attempt and retry test
REQ-LOG-010	The system shall record return-to-supplier outcomes and make the items available again only after inspection.	Must	BR-018	Returned parcel stock test
REQ-LOG-011	The system shall record carrier claims with evidence, amount, status and outcome.	Should	CAP-10	Claim lifecycle test
REQ-LOG-012	The system shall provide a customer tracking page by order and shipment without exposing internal notes.	Must	BN-11	Tracking page test
REQ-LOG-013	The system shall show an estimated delivery date derived from carrier service and handling time, labelled as an estimate.	Should	BN-06	Estimate calculation test
REQ-LOG-014	The system shall compute carrier on-time and first-attempt performance from system events.	Must	BR-030	Scorecard calculation test

ID	Requirement	Priority	Source	Verification
REQ-LOG-015	The system shall record the customer delivery charge, carrier charge and coordination margin separately.	Must	BR-005	Three amounts visible in finance report
REQ-LOG-016	The system shall produce a pickup manifest per supplier and pickup run.	Should	CAP-10	Manifest generation test

4.11 Returns, refunds and disputes (REQ-RET)

ID	Requirement	Priority	Source	Verification
REQ-RET-001	The system shall determine return eligibility from delivery date, category, item condition rules and policy version.	Must	BR-035	Eligibility matrix test
REQ-RET-002	The system shall require a return reason code and allow photo evidence.	Must	CAP-11	Request with and without evidence
REQ-RET-003	The system shall display non-returnable conditions before purchase and again at request time.	Must	BR-035	Disclosure visible
REQ-RET-004	The system shall route a request to the supplier and to support according to policy and value thresholds.	Must	CAP-11	Routing test
REQ-RET-005	The system shall support approval, rejection and partial approval with a reason.	Must	CAP-11	Decision test
REQ-RET-006	The system shall coordinate reverse pickup or drop-off and record the reverse shipment.	Must	CAP-11, CAP-10	Reverse shipment record
REQ-RET-007	The system shall require inspection before refund when the policy requires it.	Should	BR-035	Refund blocked before inspection
REQ-RET-008	The system shall calculate the refund from item value, delivery share, discounts and prior refunds.	Must	BR-006, BR-027	Calculation test
REQ-RET-009	The system shall charge the refunded amount back to the supplier’s payable where the supplier is responsible.	Must	BR-004	Supplier statement reflects chargeback
REQ-RET-010	The system shall support partial return of an order line quantity where the variant is identical.	Should	CAP-11	Partial quantity test
REQ-RET-011	The system shall convert a contested return into a dispute with evidence and a decision record.	Must	CAP-11	Dispute creation test
REQ-RET-012	The system shall escalate unresolved disputes past a configurable age to the operations lead.	Must	CAP-11	SLA escalation test
REQ-RET-013	The system shall notify the customer of each return and refund decision within the published service level.	Must	BN-10	Notification test
REQ-RET-014	The system shall close the return only when the refund, replacement or rejection is complete and communicated.	Must	CAP-11	Closure state test

4.12 Customer support (REQ-CS)

ID	Requirement	Priority	Source	Verification
REQ-CS-001	The system shall create a ticket from a customer contact with order, supplier order and shipment context where available.	Must	CAP-12	Ticket creation from order
REQ-CS-002	The system shall display order timeline, payment status, shipment events and prior tickets on one screen.	Must	BN-12	Support screen completeness
REQ-CS-003	The system shall assign tickets by queue, category and workload.	Should	CAP-12	Assignment rule test
REQ-CS-004	The system shall track first-response and resolution timers against the service level.	Must	BN-12	Timer breach flagged
REQ-CS-005	The system shall record internal notes that are never visible to customers or suppliers.	Must	ROL-07	Visibility test
REQ-CS-006	The system shall provide approved response templates in Swahili and English.	Should	CAP-12	Template insert test

ID	Requirement	Priority	Source	Verification
REQ-CS-007	The system shall route refund requests above a threshold to finance approval.	Must	BR-009	Threshold routing test
REQ-CS-008	The system shall record a resolution code for every closed ticket.	Must	BN-24	Closure requires code
REQ-CS-009	The system shall collect a satisfaction rating after resolution.	Should	CAP-12	Rating stored
REQ-CS-010	The system shall search tickets by customer, order, phone and reference.	Must	NFR-SUP-01	Search test

4.13 Notifications (REQ-NOT)

ID	Requirement	Priority	Source	Verification
REQ-NOT-001	The system shall send transactional messages for verification, order, payment, dispatch, delivery, return and refund events.	Must	CAP-13	Trigger test per event
REQ-NOT-002	The system shall support SMS, WhatsApp and email channels, with channel availability configurable per message type.	Must	CAP-13	Channel fallback test
REQ-NOT-003	The system shall use approved templates with parameters and render them in the recipient’s language.	Must	NFR-I18N-01	Template rendering test
REQ-NOT-004	The system shall send marketing messages only where consent exists and shall honour withdrawal immediately.	Must	BR-025	Consent enforcement test
REQ-NOT-005	The system shall retry failed messages and record permanent failures for follow-up.	Must	NFR-R-03	Failure retry test
REQ-NOT-006	The system shall record delivery status for each message where the provider reports it.	Should	CAP-13	Status callback test
REQ-NOT-007	The system shall allow a customer to opt out of non-essential messages without losing order updates.	Must	BR-025	Opt-out test
REQ-NOT-008	The system shall support a safety notice broadcast to affected buyers of a recalled or suspended product.	Must	BR-040	Recall notice test
REQ-NOT-009	The system shall never include full payment credentials or sensitive KYB data in a message.	Must	NFR-PRIV-02	Message content audit

4.14 Administration, access and audit (REQ-ADM)

ID	Requirement	Priority	Source	Verification
REQ-ADM-001	The system shall provide staff accounts separate from customer and supplier accounts.	Must	CAP-14	Account type separation test
REQ-ADM-002	The system shall enforce role-based permissions with deny-by-default and tenant scoping for supplier users.	Must	NFR-SEC-04	Authorisation matrix tests
REQ-ADM-003	The system shall mask personal and financial data by default and reveal it only with a permission and a reason.	Must	NFR-PRIV-02	Masking and reveal tests
REQ-ADM-004	The system shall maintain configuration for fees, commission, zones, policies, checklists, templates and brand rules with effective dates.	Must	BR-022	Configuration change without deployment
REQ-ADM-005	The system shall keep an append-only audit log of authentication, role, refund, payout, price, listing and supplier-status events.	Must	BR-023	Audit coverage test
REQ-ADM-006	The system shall support time-bound break-glass access with a reason, alerting and post-hoc review.	Should	BR-026	Break-glass grants and expires
REQ-ADM-007	The system shall support controlled data export for operations and finance with permission and logging.	Must	CAP-14	Export permission test
REQ-ADM-008	The system shall support feature flags to enable or disable capabilities without deployment.	Should	CAP-14	Flag toggle test

ID	Requirement	Priority	Source	Verification
REQ-ADM-009	The system shall support approval workflows for refunds, payouts, supplier approval and fee changes.	Must	BR-009	Approval chain test
REQ-ADM-010	The system shall invalidate all sessions for a user when access is revoked.	Must	NFR-SEC-04	Revocation test
REQ-ADM-011	The system shall record the reason for every manual status override and surface overrides in a weekly report.	Must	BN-15	Override report test
REQ-ADM-012	The system shall restrict bulk personal-data export to named roles and require a second approval above a record threshold.	Should	BR-026	Threshold approval test

4.15 Reporting and analytics (REQ-RPT)

ID	Requirement	Priority	Source	Verification
REQ-RPT-001	The system shall provide a daily operations dashboard for orders, payments, deliveries, exceptions and tickets.	Must	BN-24	Dashboard completeness check
REQ-RPT-002	The system shall report orders and gross order value by day, department, supplier and status.	Must	BN-19	Report totals match ledger and orders
REQ-RPT-003	The system shall report revenue, commission, delivery margin and contribution by department and period.	Must	BN-19	Contribution calculation test
REQ-RPT-004	The system shall report delivery performance by carrier, zone and supplier.	Must	BR-030	Performance report test
REQ-RPT-005	The system shall report returns by category, reason, supplier and outcome.	Must	BN-10	Return report test
REQ-RPT-006	The system shall provide a monthly supplier scorecard covering dispatch timeliness, cancellation, return and rating.	Must	BN-17	Scorecard calculation test
REQ-RPT-007	The system shall report payment reconciliation exceptions and ageing.	Must	BN-23	Exception ageing report
REQ-RPT-008	The system shall report customer repeat purchase and cohort retention where data volume permits.	Should	OBJ-08	Cohort report test
REQ-RPT-009	The system shall export reports as CSV with the filters applied.	Must	CAP-15	Export matches screen
REQ-RPT-010	The system shall state the data freshness timestamp on every report.	Should	BN-24	Timestamp visible

4.16 Reviews and ratings (REQ-REV)

ID	Requirement	Priority	Source	Verification
REQ-REV-001	The system shall allow a review only from a customer with a delivered purchase of that line.	Must	BR-014	Non-buyer review blocked
REQ-REV-002	The system shall allow one review per purchased order line.	Must	BR-014	Duplicate blocked
REQ-REV-003	The system shall moderate reviews for abuse and personal data, not for disagreement.	Must	BR-014	Moderation reason codes
REQ-REV-004	The system shall allow a supplier to post one public reply per review.	Should	CAP-16	Reply test
REQ-REV-005	The system shall allow any user to report a review for abuse.	Should	CAP-16	Report flow test
REQ-REV-006	The system shall compute and display an average rating only when a minimum number of reviews exists.	Should	CAP-16	Threshold display test
REQ-REV-007	The system shall not allow a review to be edited after publication except for privacy removal.	Should	BR-014	Edit blocked

4.17 Promotions and vouchers (REQ-PRO)

ID	Requirement	Priority	Source	Verification
REQ-PRO-001	The system shall allow an administrator to create a voucher with code, value, minimum basket, validity and usage limits.	Should	CAP-17	Voucher creation test
REQ-PRO-002	The system shall record the funding source of every discount: platform or supplier.	Must	BR-005	Funder visible in ledger
REQ-PRO-003	The system shall enforce eligibility and stacking rules at checkout.	Should	CAP-17	Stacking rule test
REQ-PRO-004	The system shall apply the discount to eligible lines and allocate it proportionally across suppliers.	Should	BR-004	Allocation test
REQ-PRO-005	The system shall attribute every redemption to a campaign for reporting.	Should	BN-21	Campaign report
REQ-PRO-006	The system shall enforce campaign budget caps and stop redemption when exhausted.	Could	CAP-17	Cap test
REQ-PRO-007	The system shall prevent supplier-funded discounts from reducing the supplier payable below zero without an explicit negative-balance rule.	Should	BR-029	Negative balance test

4.18 Brand and franchise configuration (REQ-FRN)

ID	Requirement	Priority	Source	Verification
REQ-FRN-001	The system shall store brand assets, trading name, contact details and policy text as configuration.	Must	BR-022	Configuration change test
REQ-FRN-002	The system shall store territory rules and display them where a territory restriction applies.	Should	BR-022	Territory rule test
REQ-FRN-003	The system shall store franchise or licence fee display rules and keep them inactive by default.	Must	BR-022, D-10	Default state is inactive
REQ-FRN-004	The system shall require a recorded licence reference before any franchise-branded configuration can be activated.	Must	D-10	Activation blocked without reference
REQ-FRN-005	The system shall record which brand configuration was active on each order for evidence.	Should	BN-20	Order brand record
REQ-FRN-006	The system shall produce a compliance report of brand and territory configuration changes.	Could	CAP-18	Report generation test

5. Use cases

UC-01 Multi-supplier purchase and payment

Actors: Customer, PSP, system. **Trigger:** customer proceeds to checkout. **Preconditions:** at least one purchasable variant in the cart.

Main flow

1. Customer opens the cart; the system revalidates price and stock per line.
2. Customer enters or selects an address with a landmark.
3. System groups lines by supplier, quotes delivery per shipment and shows item value, delivery, discounts, any small-order fee and total.
4. Customer accepts terms and privacy notice; system records consent with version.
5. System creates the parent order, supplier orders and reservations, and requests payment from the PSP.

6. Customer completes payment in the PSP interface.
7. PSP sends a signed callback; system verifies the signature and queries the PSP status.
8. System marks payment captured, confirms the order and notifies customer and suppliers.
9. Each supplier accepts or rejects within the acceptance window.

Alternates

- 4a. Cart below minimum basket: system offers to add items or applies the disclosed small-order fee.
- 7a. Callback invalid: system rejects it, records the attempt and waits for the verified status.
- 7b. Payment fails: system releases reservations after the retry window, keeps the cart and notifies the customer.
- 9a. Supplier rejects: system cancels that supplier order, initiates a partial refund for its lines and delivery share, and continues the rest.

Postconditions: parent order exists with correct payment state; each supplier order has an owner and a state; ledger entries exist for payment and payables. **Requirements:** REQ-CHK-001 to 015, REQ-ORD-001 to 009, REQ-PAY-001 to 014.

UC-02 Supplier onboarding and verification

Actors: Supplier applicant, operations, finance. **Trigger:** application submitted.

Main flow

1. Applicant submits business, owner and payout details and uploads documents.
2. System checks completeness, duplicates and document validity.
3. Operations reviews documents, conducts a site visit and records evidence.
4. Finance verifies the payout account name against the legal or trading name.
5. Operations approves, rejects or requests information, with a reason.
6. On approval, the contract version is recorded and the supplier can publish listings after moderation.

Alternates

- 2a. Duplicate TIN or bank account: application is held for review.
- 4a. Name mismatch: approval blocked until evidence is provided.
- 5a. Information requested: applicant corrects and resubmits the same record.

Postconditions: supplier state, evidence and decision are recorded and auditable. **Requirements:** REQ-KYC-001 to 014.

UC-03 Listing creation and moderation

Actors: Supplier, moderator, category lead. **Trigger:** supplier submits a listing.

Main flow

1. Supplier selects a category; system presents the attribute template and checklist.
2. Supplier enters product, variants, SKU, price, stock, media and category attributes.
3. System validates mandatory fields, SKU uniqueness, brand claims and prohibited terms.
4. Listing enters the moderation queue with the category checklist.
5. Moderator approves, rejects or requests changes with reason codes.
6. Approved and in-stock listings become visible; rejected listings return to the supplier with reasons.

Alternates

- 3a. Mandatory attribute missing: submission blocked with field-level guidance.
- 5a. Suspected counterfeit or unsafe item: listing held and escalated to the category lead.
- 6a. Stock reaches zero after approval: listing is hidden automatically.

Postconditions: listing version, decision and reason are recorded. **Requirements:** REQ-CAT-001 to 016, REQ-INV-005.

UC-04 Supplier fulfilment and dispatch

Actors: Supplier, operations, carrier. **Trigger:** supplier accepts an order.

Main flow

1. Supplier sees only its order lines and the delivery data needed to pack.
2. Supplier marks the order ready and records package count, weight and dimensions.
3. System creates shipment records and requests pickup or drop-off per the configured mode.
4. Supplier hands over parcels; waybill is recorded against each shipment.
5. Shipment status events are recorded until delivery and proof of delivery.

Alternates

- 2a. Item unavailable after acceptance: supplier requests a line cancellation with reason; system recalculates the order and refunds.
- 3a. Carrier does not arrive: operations reassigns or rebooks and records the failure.
- 4a. Waybill missing: shipment remains unbooked and is flagged for operations.

Postconditions: shipments exist with carrier, waybill and cost; supplier order reflects dispatch. **Requirements:** REQ-ORD-002 to 009, REQ-LOG-001 to 016.

UC-05 Manual logistics booking and delivery

Actors: Operations, carrier, customer. **Trigger:** shipment ready for pickup.

Main flow

1. Operations selects carrier and service from the configured register and rate rules.
2. System shows the quoted cost and the customer delivery charge separately.
3. Operations books the pickup with the carrier and records the waybill and pickup date.
4. Status events are entered or imported as the parcel moves.
5. On delivery, operations or the carrier records proof of delivery.
6. System marks the shipment delivered and starts the return window.

Alternates

- 3a. Quoted cost exceeds the charged delivery amount beyond the threshold: operations escalates before booking.
- 5a. Delivery fails: reason is recorded and a retry is scheduled per policy.
- 5b. Parcel lost or damaged: claim is opened and the customer is refunded or replaced.

Postconditions: shipment timeline and POD exist; delivery performance is measurable. **Requirements:** REQ-LOG-002 to 015, REQ-ORD-006.

UC-06 Return request and refund

Actors: Customer, support, supplier, finance. **Trigger:** customer requests a return within the window.

Main flow

1. System checks eligibility by delivery date, category and policy version.
2. Customer selects lines, quantity and reason and adds evidence.
3. System routes the request to support and, where policy allows, the supplier.
4. Decision is recorded: approved, partially approved or rejected, with reason.
5. Reverse pickup or drop-off is arranged and recorded.
6. Item is inspected where required.
7. System calculates the refund including the delivery share where applicable.
8. Finance approves above threshold; PSP processes the refund; ledger and supplier payable are updated.
9. Customer is notified and the return closes.

Alternates

- 1a. Outside window or non-returnable: request blocked with the policy explanation and a support link.
- 4a. Supplier contests: dispute is opened with evidence and escalated.
- 7a. Prior refund exists: calculation nets prior refunds and prevents over-refund.

Postconditions: refund recorded, supplier charged back where responsible, return closed with evidence. **Requirements:** REQ-RET-001 to 014, REQ-PAY-006 to 009.

UC-07 Dispute resolution

Actors: Support, operations lead, finance, supplier, customer. **Trigger:** contested return, non-delivery or payment discrepancy.

Main flow

1. Dispute is opened with category, amount, evidence and affected lines.
2. System freezes the related supplier payable to the extent of the disputed amount.
3. Operations investigates using order, shipment and payment evidence.
4. Decision is recorded with reason and approver.
5. Resolution is executed: refund, release, replacement or rejection.
6. Hold is released or converted; both parties are notified.

Alternates

- 3a. Evidence insufficient: request more from the party and extend the timer once.
- 4a. Amount above delegated limit: founder approval required.

Postconditions: dispute closed with decision and financial effect. **Requirements:** REQ-RET-011 to 014, REQ-SET-005.

UC-08 Supplier settlement and payout

Actors: Finance preparer, finance approver, supplier. **Trigger:** settlement cycle date.

Main flow

1. System calculates eligible payables: delivered, past return window, no holds.

2. System generates statements and a draft payout batch.
3. Preparer reviews exceptions and submits the batch.
4. A different approver reviews statements and exceptions and approves.
5. System exports the payout file and releases it to the channel.
6. Confirmations are reconciled; paid status and references are recorded.
7. Suppliers are notified and can view statements.

Alternates

- 1a. Open dispute or negative balance: payable is partially or fully held with a reason.
- 5a. Channel rejects a payment: item is flagged and retried in the next batch.
- 6a. Amount mismatch: exception is queued and the batch is not marked complete.

Postconditions: payout batch, approvals and reconciliation recorded. **Requirements:** REQ-SET-001 to 012.

UC-09 Recall and safety suspension

Actors: Category lead, founder, support, customer. **Trigger:** recall notice, safety report or regulator instruction.

Main flow

1. Category lead identifies affected products, variants or batches.
2. Listing is suspended immediately and removed from sale.
3. Affected orders and buyers are identified.
4. Safety notice is sent to affected buyers with instructions.
5. Returns or refunds are handled outside normal eligibility where safety requires.
6. Supplier is notified and evidence is retained.

Alternates

- 2a. Supplier disputes the suspension: listing remains suspended pending review.
- 5a. Refund exceeds normal limits: founder approval with a safety reason code.

Postconditions: product unavailable, buyers informed, evidence retained. **Requirements:** REQ-CAT-009, REQ-NOT-008, REQ-RET-001.

UC-10 Support ticket resolution

Actors: Customer, support agent, operations, finance. **Trigger:** customer contact by any channel.

Main flow

1. Agent creates or receives a ticket linked to an order where possible.
2. System shows order timeline, payment, shipment and prior contacts.
3. Agent classifies the issue and responds within the first-response target.
4. Resolution action is taken: information, refund request, return, carrier claim or escalation.
5. Resolution code is recorded and the customer is notified.
6. Satisfaction rating is requested.

Alternates

- 3a. Issue needs another team: ticket is assigned with context and SLA continues.
- 4a. Resolution needs approval: ticket waits in an approval state, not closed.

Postconditions: ticket closed with code and evidence; SLA measured. **Requirements:** REQ-CS-001 to 010.

6. User stories and acceptance criteria

Stories are grouped by actor. Acceptance criteria use Given/When/Then where the behaviour is subtle; otherwise a testable statement is given.

6.1 Customer

ID	Story	Acceptance criteria
US-01	As a customer, I want to buy items from several shops in one checkout so that I pay and track once.	Given three suppliers in the cart, when I pay, then one parent order and three supplier orders exist and I see one order with three progress rows.
US-02	As a customer, I want to know the delivery cost before paying.	Delivery is shown per shipment and in the total before the payment step; changing the address recalculates it.
US-03	As a customer, I want to see whether an item will fit before buying.	Fashion and baby listings show a size guide and size attributes; missing size blocks publication.
US-04	As a customer, I want to return an item that does not fit.	Given delivery within 7 days, when I request a return with a fit reason, then the request is accepted and routed, and the policy terms shown match the category.
US-05	As a customer, I want to know where my parcel is.	Tracking shows current shipment state, carrier, waybill and estimated date without internal notes.
US-06	As a customer, I want a real person to help when something goes wrong.	A ticket created during support hours receives a human first response within four working hours.
US-07	As a customer, I want to pay without creating an account.	Guest checkout completes with phone, name and address only, and consent is recorded.
US-08	As a customer, I want to control marketing messages.	Withdrawing marketing consent stops promotional messages and does not stop order updates.

6.2 Supplier

ID	Story	Acceptance criteria
US-09	As a supplier, I want to list my range without a laptop.	A listing with variants, price, stock and photos can be created on a phone; mandatory fields are indicated.
US-10	As a supplier, I want to know what to pack and where it goes.	The supplier order shows only its lines, recipient name, phone, address and landmark, and a packing checklist.
US-11	As a supplier, I want to be paid accurately and on time.	My statement lists orders, commission, refunds, adjustments and net payable, and matches the payout reference.
US-12	As a supplier, I want to contest a return that is not my fault.	I can add evidence and open a dispute; the payable is held only to the disputed amount.
US-13	As a supplier, I want to know how I am performing.	A monthly scorecard shows dispatch timeliness, cancellations, returns and rating, computed from system events.
US-14	As a supplier, I want clear reasons when a listing is rejected.	Rejection includes a reason code and a note, and I can correct and resubmit the same listing.

6.3 Operations and finance

ID	Story	Acceptance criteria
US-15	As an operator, I want to verify a supplier before they sell.	No listing can publish until KYC is approved, the contract is recorded and the payout name matches.
US-16	As a moderator, I want a queue that tells me what to check.	Each listing shows its category checklist, mandatory attributes and flags; decisions require a reason.
US-17	As an operator, I want to book a carrier manually and keep the same records an API would give.	Manual booking requires carrier, service, waybill, cost and pickup date, and creates the same shipment record as Mode B.
US-18	As finance, I want to reconcile payments daily.	The reconciliation run matches PSP lines to ledger entries, queues exceptions and reports unmatched value.
US-19	As finance, I want payouts approved by someone else.	A batch prepared by one user cannot be approved by the same user.
US-20	As an operator, I want to stop a dangerous product immediately.	Suspending a listing removes it from sale at once and notifies affected buyers.

6.4 Control and oversight

ID	Story	Acceptance criteria
US-21	As a founder, I want to see contribution by department.	The monthly report shows revenue, commission, delivery margin, refunds and contribution per department.
US-22	As an auditor, I want to reconstruct a refund.	For any refund, the actor, reason, approver, amount, prior refunds and PSP reference are retrievable.
US-23	As an administrator, I want to change fees without a deployment.	A commission change with an effective date applies only to orders after that date; prior orders keep their rule.
US-24	As a founder, I want to decide at month 6 with evidence.	The review pack contains orders, repeat rate, delivery performance, returns, supplier retention and contribution.

7. External interface requirements

7.1 Payment service provider

Interface	Direction	Requirement
Payment initiation	Outbound	Create a payment for the parent order with amount, currency, reference and return URL; receive a PSP reference and redirect or prompt
Callback	Inbound	Accept signed status callbacks; verify signature, timestamp and reference; respond quickly and process asynchronously
Status query	Outbound	Query by PSP reference before finalising state
Refund	Outbound	Full or partial refund by PSP reference with an idempotency key

Interface	Direction	Requirement
Settlement or payout	Outbound or file	Instruct supplier payouts in the agreed channel, or export a file for manual upload
Reconciliation file	Inbound	Daily file or API listing of transactions, fees, refunds and payouts
Error handling	Both	Timeouts, retries with backoff, dead-letter queue and alerting

The platform must support replacing the PSP without changing order, ledger or settlement logic. Provider-specific fields are isolated in an adapter.

7.2 Carrier

Operation	Requirement
Quote	Return a price and estimated transit time for a parcel and destination zone
Book	Create a booking and return a waybill and label reference
Cancel	Cancel a booking before pickup where the carrier allows
Track	Return status events with timestamps and location where available
Proof of delivery	Return recipient, timestamp, method and evidence reference
Claim	Open and track a claim for loss or damage
Manual fallback	Every operation has a manual recording path with the same fields

No carrier is assumed. Usiri, if it becomes a carrier, uses the same adapter contract as any other provider.

7.3 Messaging

SMS through a Tanzanian-capable gateway; WhatsApp through an approved business provider; email for receipts and statements. Templates are versioned per language. Delivery reports update message status. Opt-out is honoured for marketing only.

7.4 Storage and media

Product images and KYC documents are stored in private object storage. Product media is served through signed, time-limited URLs or a CDN with derivative sizes. KYC documents are never publicly addressable and access is logged.

7.5 Address and location

Address capture is text-first with area, street, landmark and optional pin. The platform must not depend on a geocoding provider for checkout to work. A pin, where provided, is stored as a hint, not as the sole address.

7.6 Reporting and export

CSV exports for orders, payments, payouts, returns and catalogue. Exports respect the filters applied and are logged. No direct database access is provided to business users.

8. Data requirements

8.1 Entity overview

Entity	Key data	Relationships
Customer	Phone, name, email, consent, status	Addresses, orders, tickets, reviews
Address	Recipient, phone, area, street, landmark, pin, instructions	Customer, orders
Supplier	Legal name, trading name, TIN, status, contract, payout account	Users, products, orders, statements
Supplier document	Type, file reference, expiry, verification status	Supplier
Category	Parent, name, attributes, checklist, return policy	Products
Product	Supplier, category, title, brand, description, status	Variants, media, reviews
Variant	SKU, option values, price, stock, barcode	Product, order items, reservations
Reservation	Variant, quantity, order reference, expiry	Variant, order
Parent order	Customer, totals, payment state, address snapshot	Supplier orders
Supplier order	Supplier, amounts, commission rule, state, payout date	Items, shipments, returns
Order item	Variant snapshot, title, price, quantity, discount, tax line	Supplier order
Shipment	Carrier, service, waybill, weight, dimensions, cost, state	Supplier order, events
Shipment event	Status, source, timestamp, raw reference	Shipment
Payment	PSP reference, amount, state, method	Parent order, refunds, ledger
Refund	Payment, amount, reason, approver, PSP reference	Payment, return
Ledger entry	Account, debit, credit, reference, period	Payment, refund, payout
Payout	Supplier, period, gross, deductions, net, state, approvals	Supplier, ledger
Return	Order item, quantity, reason, evidence, state	Supplier order, refund
Dispute	Return or order, amount, evidence, state, decision	Return, supplier order
Ticket	Customer, order, category, state, SLA, resolution code	Customer, order
Review	Order item, rating, text, moderation state	Product, supplier
Consent	Purpose, notice version, timestamp, source	Customer

Entity	Key data	Relationships
Audit record	Actor, action, entity, before, after, reason, timestamp	All
Configuration	Key, value, effective dates, approver	Fees, zones, policies, brand

8.2 Data rules

1. Order items snapshot title, SKU, variant description, unit price, discount, tax line and commission rule.
2. Money is integer TZS; rounding is applied once per calculation and the rule is recorded.
3. Personal data is masked by default in lists and exports; reveal requires permission and reason.
4. KYC documents are encrypted, access-logged and retained per legal advice.
5. Audit records are append-only; no application role can update or delete them.
6. Deletion anonymises personal fields while preserving financial and audit records.
7. Category attributes are data, not schema columns; adding a department must not require a migration of product tables.

The physical schema, indexes and retention schedule are defined in the database design document. This section defines business data obligations only.

9. Non-functional requirements

9.1 Performance and scale

ID	Requirement	Target	Verification
NFR-P-01	Customer pages load quickly on a mid-range Android phone on 4G	LCP at 75th percentile within 3.0 s	Lighthouse and field measurement
NFR-P-02	Core read APIs respond quickly under pilot load	p95 within 500 ms, excluding third-party calls	Load test
NFR-P-03	Write APIs complete predictably	p95 within 1,000 ms, excluding PSP or carrier latency	Load test and tracing
NFR-P-04	Search returns relevant results at pilot catalogue size	p95 within 800 ms at 5,000 SKUs	Query benchmark
NFR-P-05	A customer can complete checkout on a mid-range device	Median under 5 minutes including payment	Moderated usability test
NFR-P-06	Product images are sized for mobile data	No image above 250 KB on listing pages; responsive derivatives	Page-weight audit
NFR-S-01	The pilot supports bursts with headroom	100 concurrent users and 10 checkouts per minute, 5x design headroom	Load test
NFR-S-02	The schema supports growth without re-architecture	10x SKUs and 5x order volume over pilot	Design review and synthetic data test

9.2 Availability and reliability

ID	Requirement	Target	Verification
NFR-A-01	Catalogue, checkout and order APIs are available during support hours	99.5% monthly, excluding announced maintenance	Uptime monitoring
NFR-R-01	Payment and order operations survive retries without duplication	Idempotent initiation and callback; zero duplicate orders in test	Integration and chaos tests
NFR-R-02	Data can be recovered after failure	RPO 15 minutes; RTO 4 hours	Restore drill
NFR-R-03	Background jobs retry safely and surface dead letters	At-least-once delivery with idempotent handlers; alert on failures	Job failure injection

9.3 Security

ID	Requirement	Target	Verification
NFR-SEC-01	Authentication, checkout and admin meet a recognised baseline	OWASP ASVS Level 2 for in-scope controls	Security review and penetration test
NFR-SEC-02	Traffic is encrypted and secrets protected	TLS 1.2 or higher; managed secret store; no secrets in logs	Configuration review and scan
NFR-SEC-03	Passwords and staff authentication resist compromise	Argon2id; MFA for staff and supplier owners	Code review and test
NFR-SEC-04	Authorisation is enforced server-side on every request	Deny-by-default; tenant scoping; automated authorisation tests	Test suite and manual review
NFR-SEC-05	Abuse is rate-limited	Login 5 per minute per IP; OTP 3 per 10 minutes per number; checkout 10 per minute per user	Configuration test
NFR-SEC-06	Uploads cannot become an attack vector	Type and size limits; malware scan; private storage; signed access	Upload tests
NFR-SEC-07	Dependencies and images are scanned before release	No known critical vulnerability in production artefacts	CI report
NFR-SEC-08	Security headers and cookie attributes are enforced	HSTS, CSP, frame denial, SameSite cookies	Header scan

9.4 Privacy and audit

ID	Requirement	Target	Verification
NFR-PRIV-01	Consent and privacy notices are recorded and enforceable	Versioned notice; consent per purpose; withdrawal honoured	Process and data audit
NFR-PRIV-02	Personal data is minimised in logs and interfaces	No card data; PII masked by default; access logged	Log inspection and role tests

ID	Requirement	Target	Verification
NFR-PRIV-03	Data subject requests can be fulfilled	Export, correction and deletion within the statutory period to be confirmed	Tabletop exercise with counsel
NFR-AUD-01	Sensitive actions are traceable	100% of role, refund, payout, price and supplier-status changes logged with actor, reason and before/after	Audit coverage test
NFR-AUD-02	Audit records resist tampering	Append-only storage; separate access; retention per legal advice	Design review and access test

9.5 Maintainability and operability

ID	Requirement	Target	Verification
NFR-M-01	Critical modules have automated tests	At least 70% unit coverage on order, payment, commission and refund logic; integration tests for adapters	CI report
NFR-M-02	Database changes are controlled	Versioned migrations; tested rollback; no manual production schema edits	Migration review
NFR-M-03	Code and decisions are documented	Architecture decision records; API specification generated from source; runbooks	Documentation review
NFR-OBS-01	Failures are visible before customers report them	Structured logs with request IDs; error tracking; payment-success and uptime alerts; dashboards	Incident drill
NFR-OPS-01	Deployments are repeatable and reversible	One-command deploy; rollback within 15 minutes; no manual production edits	Deploy and rollback drill
NFR-SUP-01	Support can find any order quickly	Order lookup by number, phone, supplier or tracking reference in under 30 seconds	Usability test with support staff

9.6 Accessibility, localisation and compatibility

ID	Requirement	Target	Verification
NFR-ACC-01	Critical customer flows are usable with assistive technology	WCAG 2.1 AA for catalogue, cart, checkout and tracking	Accessibility audit
NFR-I18N-01	The interface works in Swahili and English	Swahili default; all strings externalised; TZS formatting; no hard-coded copy	Content review and pseudo-localisation test
NFR-COMP-01	The site works on the devices customers use	Current Chrome, Safari, Firefox and Android WebView; graceful degradation	Device lab and analytics

10. State models

10.1 Payment

From	Event	To	Authority
—	Checkout	Initiated	System
Initiated	PSP accepted	Pending	PSP
Pending	Verified capture	Captured	System after status query
Pending	Failure or expiry	Failed	System
Captured	Refund completed	Partially refunded or Refunded	Finance approval and PSP
Captured	Chargeback	Reversed or Disputed	Finance
Failed	Retry within window	Initiated	Customer

10.2 Supplier order

From	Event	To	Authority
—	Parent order paid	Awaiting acceptance	System
Awaiting acceptance	Supplier accepts	Accepted	Supplier
Awaiting acceptance	Supplier rejects or window expires	Cancelled	Supplier or system
Accepted	Supplier starts packing	Preparing	Supplier
Preparing	Package ready	Ready for pickup	Supplier
Ready for pickup	Carrier takes parcel	Dispatched	Operations or carrier event
Dispatched	POD recorded	Delivered	Carrier or operations
Delivered	Return window closes	Completed	System
Any pre-delivery	Cancellation approved	Cancelled	Supplier or operations, with reason

10.3 Shipment

Draft → Booked → Picked up → In transit → Out for delivery → Delivered. Exceptions: Failed attempt → Retry scheduled or Returned to supplier; Lost; Damaged. Only carrier events or an authorised operator can move a shipment to Delivered, and a POD is mandatory.

10.4 Return

Requested → Under review → Approved or Rejected → In reverse transit → Received → Inspected → Refunded or Partially refunded → Closed. A contested decision moves to Disputed and may return to Approved or remain Rejected.

10.5 Dispute

Opened → Under review → Resolved for customer, Resolved for supplier, or Escalated → Closed. A dispute freezes the disputed amount of the supplier payable until closed.

10.6 Customer-facing summary

The summary is derived, never typed: **Processing** while any supplier order is awaiting acceptance or preparing; **Shipped** when all active supplier orders are dispatched; **Partially delivered** when some shipments are delivered and others are not; **Delivered** when all shipments are delivered; **Cancelled** when all supplier orders are cancelled; **Return in progress** when any return is open; **Refunded** when refunds cover the captured amount. The customer sees both the summary and the per-supplier detail.

10.7 Transition authority

Transition	Who may perform	Evidence required
Mark supplier order accepted or rejected	Supplier owner or staff with permission	Reason for rejection
Mark ready for pickup	Supplier	Package count and weight
Mark dispatched	Operations or carrier event	Waybill
Mark delivered	Carrier event or operations	POD
Approve refund	Support within limit; finance above	Reason and approver
Release payout	Finance approver, not preparer	Batch review
Suspend listing or supplier	Moderator, category lead or operations lead	Reason code
Override a state	Operations lead only	Reason, logged and reported

11. Business rule enforcement

Rule group	Enforced in	Evidence
Order structure and snapshots (BR-001, BR-011)	Order service and database constraints	Order record and tests
Server-side pricing (BR-003)	Checkout calculation service	Tampering tests
Commission on net value (BR-004)	Settlement service	Calculation tests and statements
Separate money amounts (BR-005)	Ledger accounts and reports	Finance report
Refund limits (BR-027)	Refund service	Over-refund test
Payout holds and maker-checker (BR-007, BR-028, BR-029)	Settlement workflow	Approval tests
Catalogue visibility (BR-010, BR-015, BR-034)	Catalogue service and moderation workflow	Publish-block tests
Delivery evidence (BR-008, BR-016)	Shipment service	POD requirement
Return eligibility (BR-018, BR-035)	Returns service with policy configuration	Eligibility matrix tests

Rule group	Enforced in	Evidence
Variants and SKU (BR-033)	Catalogue and inventory	Variant checkout tests
Access and audit (BR-023, BR-026)	Authorisation layer and audit service	Authorisation and audit tests
Brand and franchise (BR-022)	Configuration service	Activation and effective-date tests

A rule that cannot be enforced in software must be enforced by a documented operating control, and that control must be named in the test plan. “We will remember” is not a control.

12. Traceability

12.1 Objective to requirement

Objective	Business needs	Requirement modules	Acceptance theme
OBJ-01 Demand	BN-01, BN-04, BN-05	REQ-CHK, REQ-ORD, REQ-DIS, REQ-INV	Three-supplier checkout completes and is tracked as one order
OBJ-02 Delivery	BN-06, BN-11	REQ-LOG, REQ-ORD, REQ-NOT	Every shipment has carrier, waybill, events and POD
OBJ-03 Trust	BN-09, BN-10, BN-12	REQ-RET, REQ-CS, REQ-PAY	Return resolved with refund, chargeback and communication
OBJ-04 Supply	BN-02, BN-03, BN-17	REQ-KYC, REQ-CAT, REQ-RPT	Verified supplier publishes approved variants and sees own data only
OBJ-05 Economics	BN-08, BN-19, BN-23	REQ-SET, REQ-PAY, REQ-RPT	Commission, refunds and payouts reconcile to the ledger
OBJ-06 Compliance	BN-13, BN-15, BN-26	REQ-ACC, REQ-ADM, REQ-NOT	Consent, terms version, invoices and audit are evidenced
OBJ-07 Operability	BN-12, BN-24, BN-27	REQ-CS, REQ-RPT, REQ-ADM	A pilot day runs in the system without shadow spreadsheets
OBJ-08 Scale decision	BN-28, BN-25	REQ-RPT	Month-6 pack shows demand, retention, delivery and contribution

12.2 Rules of traceability

1. Every Must requirement references at least one business need and one verification method.
2. Every business rule marked Confirmed or Proposed has at least one requirement that enforces it or a named manual control.
3. Every test case references a requirement ID; a test without a requirement is either unnecessary or exposes a missing requirement.
4. The traceability matrix is a living file, updated in the same change as the requirement.
5. No requirement is deleted; it is marked withdrawn with a reason and a replacement reference.

13. Verification and acceptance

13.1 Test levels

Level	Scope	Owner
Unit	Calculation, state transitions, validation	Engineering
Integration	PSP sandbox, carrier adapter stub, messaging, storage	Engineering
End-to-end	Customer, supplier, operations journeys on staging	QA and operations
Security	Authorisation matrix, input handling, callback forgery, dependency scan	Technical lead and reviewer
Performance	Load and soak at pilot targets with headroom	Engineering
Accessibility	WCAG 2.1 AA on critical flows	Design and QA
User acceptance	Real operators run scripted and exploratory pilot days	Founders and operations
Pilot acceptance	Live orders measured against objectives	Founders

13.2 Entry and exit criteria

Entry to system testing: requirements baseline approved; environments available; PSP sandbox and carrier contract confirmed; test data prepared; no open critical defect.

Exit to pilot: all Must requirements verified; no open critical or high defect; reconciliation proven with seeded mismatches; restore drill passed; security review closed; legal memo received; supplier cohort and carrier ready; runbooks written.

13.3 Acceptance scenarios

ID	Scenario	Primary requirements
AT-01	A three-supplier order is paid, partially rejected, partially delivered and partially refunded, and the ledger balances.	REQ-CHK, REQ-ORD, REQ-PAY, REQ-SET
AT-02	A supplier is rejected for a name mismatch, corrects the record and is approved without duplication.	REQ-KYC-004, REQ-KYC-008, REQ-KYC-013
AT-03	A fashion listing with size variants is rejected for a missing size guide, corrected and published.	REQ-CAT-002, REQ-CAT-013, REQ-CAT-008
AT-04	A payment callback is forged and rejected; the genuine callback is verified and the order proceeds once.	REQ-PAY-002, REQ-PAY-003, REQ-PAY-004
AT-05	A delivery fails twice, is returned to the supplier and the customer is refunded with the correct delivery treatment.	REQ-LOG-009, REQ-LOG-010, REQ-RET-008
AT-06	A beauty listing is reported as expired; the listing is suspended and affected buyers are notified.	REQ-CAT-009, REQ-NOT-008, REQ-RET-001
AT-07	A payout batch prepared by finance cannot be approved by the same person.	REQ-SET-006

ID	Scenario	Primary requirements
AT-08	A customer requests deletion; personal fields are anonymised and financial records remain.	REQ-ACC-010
AT-09	A support agent resolves a ticket within SLA using only the console.	REQ-CS-001, REQ-CS-002, REQ-CS-004
AT-10	The daily reconciliation identifies a seeded amount mismatch and queues it.	REQ-PAY-010, REQ-PAY-011

14. Open issues

ID	Issue	Impact	Owner	Needed by
D-13	Exact department tree	Category configuration and templates	Founders	BRD sign-off
D-14	Minimum basket or small-order fee	Checkout validation and conversion	Founders	Checkout freeze
D-15	Category commission schedule	Settlement rules and supplier contracts	Founders and finance	Supplier contracts
D-16	Beauty authenticity evidence	Listing requirements and moderation	Category lead	Beauty go-live
D-17	Baby safety declaration	Listing requirements and liability	Founders	Baby go-live
D-18	Fashion return policy	Return eligibility and refunds	Founders	Policy publication
D-19	Launch supplier cohort	Launch readiness	Founders	Recruitment plan
D-20	Timeline fallback	Roadmap and scope gate	Founders and CTO	Roadmap sign-off
O-03	PSP selection and contract	Payment adapter and settlement	Finance lead	Before payment build
O-04	Carrier selection and terms	Logistics adapter and rates	Operations lead	Before logistics build
O-09	Tax treatment, invoicing and withholding	Receipts, statements, VAT lines	Accountant and counsel	Before launch
O-14	Hosting region and data location	Architecture and compliance	CTO and counsel	Before architecture sign-off
O-10	PDPA registration, DPO, retention and cross-border position	Deletion, archive and transfer design	Counsel	Before launch

A TBC value in this specification is not a defect. It is a controlled unknown that must be resolved before the related requirement is built, and the resolution must be recorded in the decision log.

Appendix A. Requirement index

Module	Prefix	Count	Primary business needs
Identity and consent	REQ-ACC	12	BN-13, BN-14
Supplier onboarding	REQ-KYC	14	BN-02, BN-17
Catalogue and moderation	REQ-CAT	16	BN-03, BN-16, BN-18
Inventory	REQ-INV	9	BN-05
Discovery	REQ-DIS	10	BN-04, BN-26
Cart and checkout	REQ-CHK	15	BN-01, BN-06
Orders	REQ-ORD	14	BN-01, BN-09, BN-11
Payments	REQ-PAY	14	BN-07, BN-23
Settlement	REQ-SET	12	BN-08, BN-23
Logistics	REQ-LOG	16	BN-06, BN-11
Returns and disputes	REQ-RET	14	BN-09, BN-10
Support	REQ-CS	10	BN-12
Notifications	REQ-NOT	9	BN-11, BN-18
Administration	REQ-ADM	12	BN-14, BN-15
Reporting	REQ-RPT	10	BN-19, BN-24, BN-28
Reviews	REQ-REV	7	BN-22
Promotions	REQ-PRO	7	BN-21
Brand and franchise	REQ-FRN	6	BN-20

Total: 207 requirements, of which the Must set forms the launch scope.

Appendix B. Glossary

Terms are as defined in the BRD, with these software additions:

Term	Meaning
Idempotency key	A client or system reference that makes a repeated operation return the original result instead of repeating the effect
Dead letter	A failed background job set aside for inspection after retries
Derived state	A state computed from events rather than stored as an editable field
Adapter	An isolated integration component that translates between the platform and an external provider
Tenant scoping	Restricting every query and action to the organisation a user belongs to

Term	Meaning
Snapshot	A stored copy of values at transaction time, unaffected by later edits
Effective dating	A configuration rule that applies from a date and leaves history unchanged
Break-glass	Temporary elevated access with a reason, alerting and review